

Systematic Review of Discussion Forums in Massive Open Online Courses (MOOCs)

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Abstract—Discussion forums are the primary and most widely used venue for social learning and communication in MOOC context. There has been an interest in the last few years to study the interactions and participation behavior in this environment. This systematic review studied eighty-four papers published between 2013–2017 to provide a state of the art, and suggest future directions in the literature from two perspectives: descriptive analysis and content analysis. The descriptive analysis aggregates the reviewed papers into five dimensions: time of the publication, stakeholders involved in the analysis, objectives, methods, and data sources used in the studies. The content analysis aims to highlight the main issues addressed and the major contributions of the papers covered by the literature review. These contributions can be categorized into three major areas, first, explore the relationship between forums participation and retention or learning outcomes and evaluate ways to increase the participation activity and learning outcomes, second, analyze participants' contribution in the forum and how to organize the content and efficiently monitor them, finally, analyze participants' interactions and how it influences learning. At the end, we offer some suggestions to further advance this area of research.

Index Terms—Discussion forums, MOOC, learning analytics, systematic review.

I. INTRODUCTION

IT has been a decade since the term Massive Open Online Courses or MOOCs was first introduced. Since its inception, the number of registrants, courses offered, and platforms available for MOOCs have been growing substantially. In 2016, more than 35 million participants signed up for at least one MOOC, and more than 4,000 courses were offered worldwide by hundreds of universities [1]. The success of MOOCs is credited to their ability to provide high quality, low-cost educational content on a large scale more efficiently [2]. In addition to providing easy access for learners, MOOCs have also attracted researchers from various disciplines such as computer science, machine learning, education, pedagogy, and statistics to explore this learning environment. MOOCs have become a fertile ground to test existing theories and uncover novel dynamics related to learning.

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Individuals who are interested in learning a topic can simply join a course of interest online for free. The easy access to content yields a large number of enrollees with different knowledge backgrounds, interests, and habits, which subsequently inhibits desired learning gains for learners and also poses challenges for learners and instructors. Within MOOC platforms, one common element is a discussion forum. A discussion forum provides participants the ability to discuss—pose questions, receive answers, and post answers; discussion forums are the primary and most widely used venue for social and learning interactions among MOOC learners and between learners and instructors [3]. Instructors perceive it as the most useful tool to monitor the dynamics of the course and assist in preparing a more developed iteration of the course [4]. Learners, on the other hand, use it to discuss content, express opinions, report problems with the course or the platform, connect with fellow participants, or form groups [5]. Given the large number of participants in MOOCs, the diversity of topics discussed and the variety of participants' motivation in MOOC forums often create an overwhelming and disorganized environment. As a result, MOOC designers and researchers alike have looked carefully at ways to mitigate the problems within a discussion forum in MOOCs. The research on this topic is scattered in different disciplines and conducted for different purposes. A lack of clarity about how MOOC discussion forums have been studied and for what purposes [6] is the drive of this paper, along with the importance of accumulating the harvests of literature and identifying the gaps for future research, especially as MOOCs have reached a milestone (a decade) since their inception.

Given the central nature of a discussion forum to interaction on MOOCs, this is a critical area that needs to be understood, designed, and investigated better. The aim of this paper is to provide a comprehensive, systematic review on discussion forums in MOOCs. In the context of MOOCs, there are few papers that have adopted a literature review approach to summarize a specific aspect of MOOCs. One of the earliest relevant literature reviews [7] identified eight themes apparent in MOOC scholarship. These include: 1) introductory articles (explaining MOOC aspects), 2) concept (debating threats and opportunities of MOOCs in higher education), 3) case studies (examining MOOC courses), 4) educational theory (examining pedagogical approaches used), 5) technology (discussing hardware and software used), 6) participant focus (investigating participants' experiences), 7) provider focus (discussing

topics related to course creators and policy makers), and 8) other [7]. Ebben and Murphy [8] traced the pedagogical development of MOOCs in its first 5 years (2009-2013). They identified two phases and differentiate between them in detail. Phase 1, cMOOC, is based on connectivist learning theory, and phase 2, xMOOC, is based more in cognitive behaviorist pedagogies. Veletsianos & Shepherdson [6] reported the geographic distribution, publication outlets, citations, data collection, analysis methods, and research strands of empirical research focusing on MOOCs in 2013–2015. The goal was to enable researchers to make more sense of the empirical literature on MOOCs and its direction and limitations. Clark *et al.* [9] surveyed peer-reviewed journal articles on MOOCs during the period of 2011-2016 to study the benefits, challenges, and students' and instructors' perceptions to describe the implications for medical imaging education. Moissa *et al.* [10] surveyed publications on learning analytics from 2011-2014 and found that less than 10 percent of the publications had studied MOOCs through the lens of learning analytics.

This paper adopts a learning analytics and content analysis perspective and has three objectives: 1) offer a comprehensive analytical overview of the existing research in the field of discussion forums in MOOCs; 2) carry out a content analysis of the main topics covered in the literature in this context; and 3) identify research gaps and potential opportunities for future direction. Overall, the review will inform researchers as well as other MOOC stakeholders about the current state of knowledge in the existing literature and the possible future directions that would help the advancement of MOOCs and the use of discussion forums in MOOCs.

II. METHODOLOGY

In this paper, a systematic review approach was adopted to ensure a comprehensive, reproducible coverage of the literature. Petticrew and Roberts [11, p. 19] defined a systematic review as a “review that strives to comprehensively identify, appraise and synthesize all relevant studies on a given topic”. In developing fields, a systematic review usually serves the purpose of describing the state of knowledge, tracing historical development, or identifying opportunities for future research [12]. The purpose of this research is to synthesize what is currently done in the literature about MOOC discussion forums to lower the barrier for future researchers and inform practitioners.

A two-step process was carried out to conduct this systematic review as in [13]. The first step was to select the relevant papers, and the second step was to synthesize the selected papers. The latter step is divided further into two sub-steps: descriptive analysis (Section III-A) and content analysis (Section III-B).

In the first step, the sources for the search, the search terms, and the inclusion and exclusion criteria were set. Five databases from the education and technology fields were used to search for relevant papers: Web of Science, Science Direct, ERIC, ACM, and IEEE Xplore. The search terms were identified at the beginning as follows:

TABLE I
THE RESULT OF APPLYING THE SEARCH TERMS IN THE DIFFERENT DATABASES AND THE NUMBER OF RELEVANT ARTICLES

Database	# Hits	# Related Articles
ERIC	11	6
Web of Science	71	28
IEEE	15	11
ACM	66–“communication of ACM”= 57	20
Science Direct	7	4
Citation search		36

- First concept: “MOOC*” OR “Massiv* Open Online Course*” AND
- Second concept: “forum*” OR “discussion*” OR “text*” OR “Message*” OR “thread*” OR “Post*” OR “Communication” OR “Interaction” OR “discourse”.

The search for these concepts was limited to the title field to maintain a manageable range of analysis. At the same time, this review has covered as much as possible of the existing literature. It includes all articles published in either conference proceedings or journals since the beginning of MOOC related papers in 2008 until 2017.

The review was limited to English articles with length of at least 4 pages that were directly related to MOOC discussion forums. The result of the search as of January 15, 2018 is presented in Table I. Table I shows the number of hits and the number of relevant articles for each database after reading the abstract and evaluating the paper based on the inclusion criteria. Some articles appeared in more than one database, and a few articles were developed further, revised, and re-published. In this case, only the more developed version of the article was included in the review.

As a next step, a backward citation search was conducted to obtain a broader coverage. After removing the duplicate work, the total number of articles included in this review amounted to 84 articles [references marked with ** are secondary articles for the paper and are not included in the descriptive or content analysis]. About 70 percent are eight and more pages. These articles include workshop proceedings and lecture notes (less than 10 percent of the total number of selected articles). Journal articles, and conference papers account for about 30 and 60 percent respectively.

In the second step, the learning analytics reference model (LARM) proposed by Chatti *et al.* [14] was used to motivate the descriptive analysis. LARM consists of four components: data, stakeholders, objective, and method. These dimensions in addition to covering the distribution of publications over time resulted in the following sub-questions for the analysis:

- What is the distribution of publications regarding MOOC forums over time?
- What are the types of data used for studying MOOC discussion forums (data)?
- Who benefited most from the research (stakeholders)?
- Why have the discussion forums been studied (objectives)?
- How did they approach the goals (methods)?

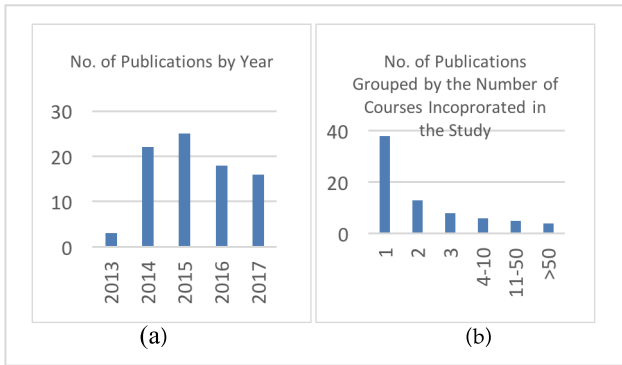


Fig. 1. Number of publications (a) by year and (b) by number of courses used in the publication.

For detailed characteristics about the four dimensions in the coding schema, and how it was used to appraise the literature and guide the process of classifying the selected papers, see Appendix A, which can be found on the IEEE Xplore Digital Library at <https://ieeexplore.ieee.org/document/8418792/>.

III. ANALYSIS

A. Descriptive Analysis

The selected papers are aggregated and mapped to the learning analytics reference model to give a summary of the current state of the art in this field.

1) Publication Distribution Over Time: In early MOOCs, known as cMOOCs, knowledge was distributed in several places. There was no single space for learners' communication and knowledge sharing. Learners used social media platform, blogs, and wiki to construct knowledge and connect with each other. In 2012, when xMOOC emerged, course content and activities mainly took place in a single platform and discussion forums were the primary channel for communications and social learning. Therefore, the topic of discussion forums in a MOOC context attracted researchers' attention later, in 2013. Then, quickly gathered momentum, peaking in 2015. After that the publications in this particular topic remained between 15-20 articles in 2016 and 2017 [See Fig. 1a].

Fig. 2 shows the co-authorship network for the selected papers of MOOC discussion forums. Nodes represent authors and links between nodes represent co-authorship between the two authors. The network consists of 234 authors and 581 links. The link thickness indicates the number of times the two authors co-author a paper, and the size of the text indicates the number of articles the author published in this topic.

2) Summarization of Literature Based on LARM: What?

There are various MOOC platforms including Coursera, edX, FutureLearn, XuetangX and MiriadaX. In the reviewed literature, 45 percent of the papers conducted studies on a single course, as illustrated in Fig. 1b. Moreover, the collected data were largely gathered from a single platform. Coursera is a leading platform in this research endeavor; about half of the papers reviewed gathered data from Coursera, and around (17 percent) used data from courses offered on edX. Additionally, around

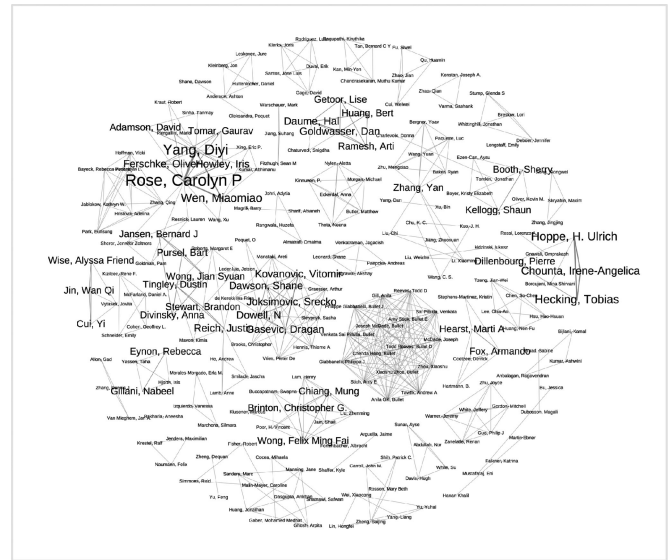


Fig. 2. Co-authorship network.

5 percent of the papers used data from multiple platforms, and 15 percent of the publications did not specify the platform used for the study or did not use data from a MOOC platform.

The data collected for analysis was organized into six groups: user-activity, posts, threads, course-info, performance data, and data collected from surveys or interviews. Fig. 3 shows the reviewed papers categorized by the type of data utilized in the studies. In many cases, more than one source is used together to achieve the goal intended for the study [Fig. 3b].

User-activity and posts are the most used data sources in the reviewed literature (each one was incorporated in more than 60 percent of the reviewed papers). Posts enclose the user-generated text and may include any of the associated information such as the time of the post, the number of up-votes and views. User-activity involves the interaction between the users of the posts and the events the user takes, such as #votes, #posts, and #replies, or who replied to whom. User activity and posts were jointly used to monitor and predict the learning in the discussion forums and to facilitate the ranking of relevant posts.

User-activity and performance combined have been used in 12 percent of the studies and more than 22 percent with other data sources. This combination of data supports examining the relationship between the engagement and learning performance, and the impact of engagement in the discussion forums on the performance. The performance data can be any of the following: certification, course completion, final grade, quiz score, number of assignments completed, or score.

Data collected using traditional methods like surveys and interviews were used (12 percent) to gain insights about learner demographics, motivation, attitude towards the discussion forums or course pedagogy, or to assess learner knowledge background and gain. Course information such as the syllabus, course content, and course landmarks were used (in around 5 percent of the reviewed papers) to map the user-generated content with the course content, to assist in

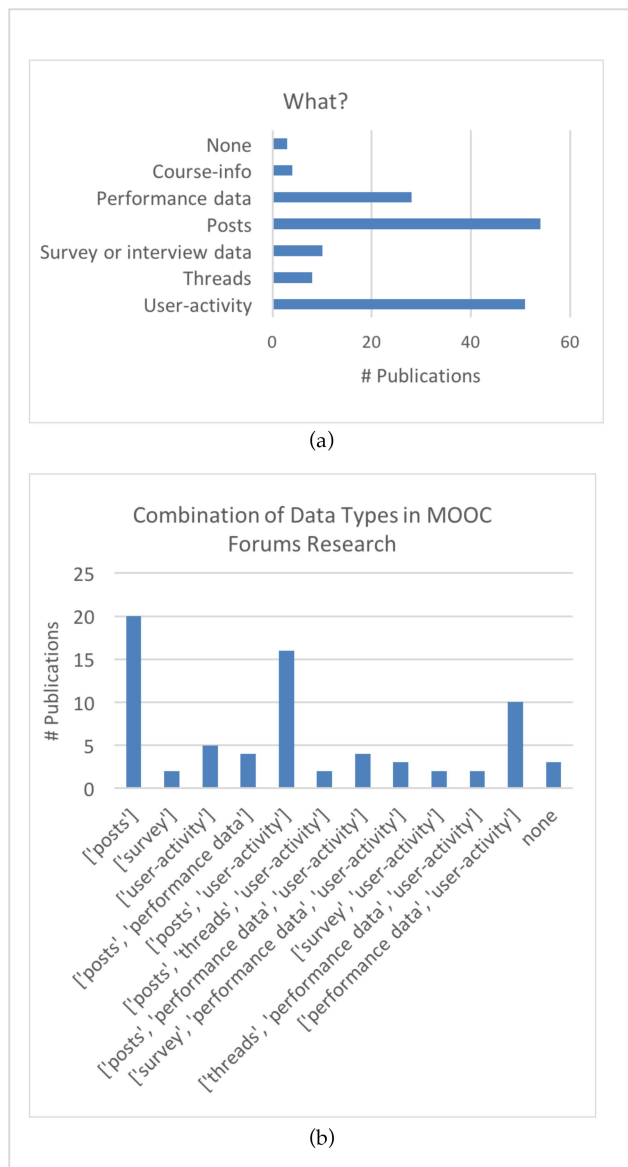


Fig. 3. The number of publications grouped by (a) data sources and (b) the combination of the data sources.

organizing the posts according to the course syllabi, or to forecast busy periods of activity.

Who?

As for the directly benefitted stakeholders for the reviewed studies, Fig. 4 shows the result. The vast majority of the research was targeted at researchers (approx. 64 percent), which might help other stakeholders as well. These studies added to the knowledge about the discussion forums in MOOC context and the interaction and knowledge creation that occur. In addition, the studies investigated the factors that affect participants' learning performance. Instructional staff, which includes course designers, instructors, teaching assistant, or facilitators, comes in second place (40 percent), assisting them to gain insights regarding the course content and teaching practices. This will help them adjust the course goals based on learners' expectations and attitude. There are few articles that directly benefit learners by recommending

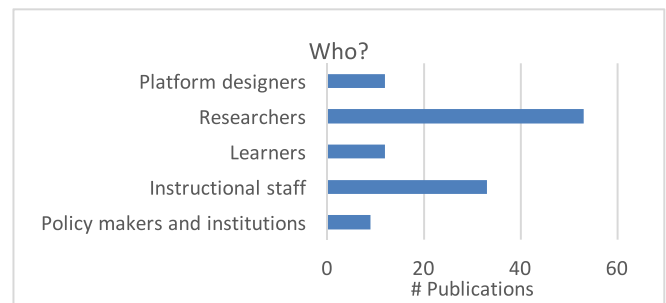


Fig. 4. The number of publications grouped by the targeted stakeholder.

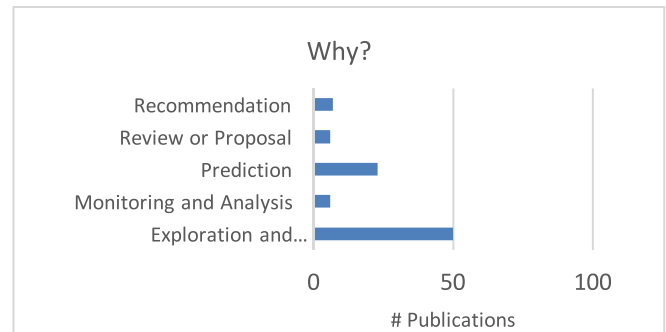


Fig. 5. The number of publications grouped by the objective of the research.

resources or peers that can help them overcome their struggles. There are also approximately 10 percent that serve policy makers and institutions by investigating the role of MOOC discussion forums in fostering learning and knowledge construction on a large-scale. Platform designers are another important stakeholder group in this research area, and about 14 percent of the articles aimed to inform them about current and potential design features and their effectiveness on learner participation patterns and learning performance.

Why?

There are different objectives for studying discussion forums in MOOCs, which ultimately aim to understand and enhance the learning experience for learners and to help instructional staff improve the orchestration of the MOOCs. The major objectives are either exploring and understanding the dynamics in discussion forums, or developing a model that automatically monitors, organizes, predicts, or recommends to learners or instructional staff. Hence, having such a model will maximize the gain out of MOOC forums. About half of the papers aimed for exploration and the other half for developing models for various objectives. Some papers aspire to do both, where they developed a model and, for instance, explored the best technique for the problem under study. Few papers (7 percent) reviewed other papers or proposed a framework to set a foundation for analyzing the data. Fig. 5 shows the reviewed papers categorized based on the intended objective.

Papers that aim for exploration and understanding mainly studied the forums and the participation patterns among participants, how they interact and learn in a collaborative learning environment, the roles participants play in the forum, and their link to the formation and health of the online community. In

addition, researchers explored the types of messages and themes that participants engage with and how participants' role and topics change over time. The papers reviewed in this category also were concerned with the learner's and instructor's experiences and needs regarding the forums and how these can be improved. Moreover, a sizable amount of the reviewed studies explored the factors that can potentially increase participation and encourage meaningful discussion, or that can evaluate the effectiveness of a pedagogy practice (e.g., instructor intervention) or a design feature on learner's participation activity or learning outcome. Furthermore, exploration studies inform researchers about the best data engineering approach or statistical technique in their studies.

Monitoring is concerned with assessing the posts to extract the topics discussed (e.g., misconceptions) or the interaction that occurred (i.e., thread reply structure). This type of research is usually augmented by visualization. A more developed research would combine monitoring with prediction referred to as prediction. There is about 28 percent of the publication in the reviewed literature fall into this category. Examples include developing models that can identify potential community teaching assistants, predict learners at risk, predict threads that are resolved [15] or need intervention, identify the cognitive level, confusion, affection, and the topic/forum that the post best belongs to [16]. These models help organize the content and allow learners and instructional staff to navigate the forums easily to find posts of interest. Many of the reviewed models in this category, however, were targeted at instructors to maximize their effectiveness and help them prioritize their intervention.

Monitoring and prediction is a pre-step for intervention and personalization, which was an objective in less than 10 percent of the reviewed articles. For example, once the topic of the post has been identified (e.g., expressed confusion) an intervention can take place. Recommendation is a form of automatic intervention. It directs participants to threads or other participants that are likely to match their interest. Personalization also assists in presenting the result in a way that best matches the user's interest and minimizes the overload on the learner's side. For example, the resources can be ranked based on the participant's preference and knowledge.

How?

There are largely six methods that have been applied in the literature, which are observation, statistics, qualitative data, data mining, social network analysis and visualization. Fig. 6 shows the reviewed literature grouped by the method used in the studies. Note that some studies applied more than one method.

Statistics was used heavily to analyze the data in the reviewed papers (42 percent of the publications). Statistics was utilized to describe the data using central metrics such as percentages or means or to find correlation and relationship between variables. In this category survival analysis (to measure dropout) and regression coefficients were widely used. Furthermore, statistics were used to test hypothesis.

Data mining was the most commonly applied method (approx. 47 percent of the reviewed papers). Data mining

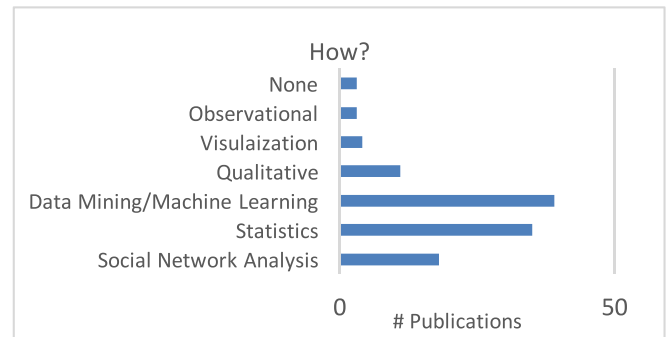


Fig. 6. The number of publications for each method category.

aims to uncover hidden patterns in large data. Classification and prediction are the most widely used approach in the data mining category. Techniques such as Naïve Bayes, regression models, support vector machine, random forest, and neural network were used to predict engagement level or learning outcome, and to classify and code messages or a participant's role. Probabilistic soft logic, which is a method that captures domain knowledge and dependencies in the domain, was applied to model learner's engagement. Content analysis, topic and feature extraction were conducted using bag-of-words, Latent Dirichlet Allocation (LDA), structural topic modelling or tools such as LIWC, Coh-metrix, and Opinion-Finder. Clustering is another technique to group similar examples together (e.g., cluster posts with similar topics [17], or participants with similar interest). Recommendation using adaptive matrix factorization, jump-random-walk, and HITS were used to recommend participants or threads. Other techniques that were infrequent include: frequent pattern mining [18], ontology [19], optimization [20], and deep learning techniques [21].

Social network analysis (SNA) considers participants or topics as nodes or actors in a graph and draws the relationships among them. It was employed in approximately 21 percent of the reviewed papers; it allows one to quantify the importance of the actors and detect clusters of actors that are more connected among each other than the random average. There were two main approaches that analyzed the SNA in MOOC discussion forums. The first approach is structural network analysis, which is mainly identified by its centrality measures such as degree, closeness, and betweenness. It was applied to study the relation between an actor position and the performance, persistence or the quality of contribution. Furthermore, it was applied to identify influential actors and sub-communities. In addition to centrality measures, blockmodeling was used to reduce graphs and discover participants' roles in the discussion forums. The second approach is statistical network analysis, a well-known method in this approach is exponential random graph modeling. It was used to estimate the likelihood of a given network structure (e.g., predict network ties) to occur for a reason other than chance.

Visualization was used in less than 5 percent of the reviewed papers. The purpose is to present the result to the targeted stakeholder in an efficient manner using charts and graphics. There are few papers that developed a dash-board or

word cloud visualization tools to maximize the instructor's efficiency in monitoring the discussion forums. In addition, there was a paper that showed the influence of different visualization of a badge system on motivating learners to participate and engage in the discussion.

B. Content Analysis

Mapping without synthesis would not provide a comprehensive view of the topic. Synthesis in a systematic review involves "pooling and exploring the results to provide a 'bottom-line' statement regarding what the evidence supports and what gaps remain in our current understanding" [22, p. 950]. In this content analysis, the aim is to highlight the main issues addressed and the major contributions covered by the literature on MOOC discussion forums. For each article, the main topic(s), result(s), and gap(s) were reported. Examining the topics, we found that articles were most likely focusing in one or more of the of the three major topics: 1) explore the relationship between forums participation and retention or learning outcomes, and evaluate ways to increase the participation activity and learning outcomes; 2) analyze participants' contribution in the forum and how to organize the content and efficiently monitor them; and 3) analyze participants' interactions and how they influence learning.

The analysis will start with a description about the participants and the participation activity in the forums and then delve into the major areas discussed earlier.

1) Participants and the Participation Activity: The participants in MOOC discussion forums can be any enrollee of the course in which the forum is embedded. Participants have different goals, motivations, and backgrounds. As MOOC becomes more established, it predominantly attracts individuals who strive to enhance their professional work skills and fulfill personal intellectual curiosity [23], [24]. This supports the demographics of the largest group of participants, in that they are employed and hold a higher education degree [25], [26]. Since many MOOCs do not require any previous knowledge, forum participants demonstrate a large spectrum of subject knowledge ranging from novice to expert. The existence of this kind of mixture is healthy for sustaining a robust ecosystem [27]. Moreover, the diversity of background knowledge enhances scaffolding, which is an important characteristic in an online learning community. It is observed when participants assist one another in building and understanding new knowledge based on prior knowledge [27].

The range of active participants varies depending on the course design (some courses require forum participation), the platform used, or the way the participation rate is calculated (based on the total number of enrollees, or the ones who accessed at least one course resource). Typically, 5-25 percent of the registered learners post at least once in the forums [26], [27], [28], [29], [30]. Furthermore, many learners choose to use the forums passively. Mustafaraj & Bu [32] observed from the event tracking database that there are two passive learners for every active one.

Previous work indicates conflicting views about learning in MOOC discussion forums and whether it resembles a community or a crowd. In a crowd, individuals start off enthusiastically with a high volume of participation, but as time passes their participation decreases [25]. The nature of communication in MOOC forums reveals several characteristics. For example, some posts get no response [28], [30], [32]; posts in the forums are not cohesive [25]; most pair communication happens only once; and participants' activity in the forums decays dramatically during the first 3 weeks, and then almost stabilizes until the end of the course [29], [32], [33], [34]. These characteristics suggest that a MOOC forum does not form an online community [29], [36]. On the other hand, [37] claims that communication in MOOC forums constitutes a community and did not evolve arbitrarily. More precisely, the study referred to the core/active participants who form social relations and demonstrate collaboration among themselves. This claim was also supported by [34], [38], [39], who found that the network (total number of participants) shrinks over time; nonetheless, the density (relationships between participants) increases. Also, [40] stated that MOOC forums meet Herring's criteria for considering an online group to be a community.

Course structure and events impact the level and nature of activity within the forums [41]. The topics of the discussions change over the course period [29]. As described by [30], early stages tend to show more social type of discourse (setting goals, expectations, introductions, and forming study groups). Then, the discourse shifts to the course materials and displays cognitive engagement. In the later stages, the discourse reverts to social/emotional purposes, mainly expressing attitudes towards the course. In terms of affective processes, the course starts with an amplified amount of positive terms that decrease over time, while negative emotions increase as the course progresses [42]. In terms of cognitive processes, they appear more frequently as the course advances [42]. Also, content related forums exhibit more cohesiveness and persistence than other types of forums [25], [36].

A recent study showed that MOOC participants and their discourse patterns are evolving in MOOC discussion forums [43]. Particularly, there were more on-topic messages in recent courses that reflected learning compared to similar earlier courses.

2) The Relationship between Participation (Behavior and Factors) and (Retention and Performance): Many studies were focused on analyzing a special group of participants, loosely called active participants. The description of active participants, however, was further refined in each study. In some cases, a smaller group of the active participants was selected for the analysis, such as active participation, regular participants, or super-poster. Table II recaps the assorted groups of participants and the major findings from the analysis.

Furthermore, participants can be categorized based on the role of the participants or the engagement behavior. Based on the role, participants can be core users, help seekers, and help givers [37]. Based on the engagement behavior, participants

TABLE II
THE DIFFERENT GROUPS OF ACTIVE PARTICIPANT, THEIR DEFINITIONS, AND
THE MAJOR FINDING ABOUT THEM

Type	Operationaliza- tion	Finding
Core posters	Participate in three weeks and re-ceived at least a response to their post.	More likely to form so- cial communities and demonstrate higher level of collaborative knowledge construction [37]
Regular participants	Show frequent presence on the fo- rum	Their activity grew over time, and they tend to have a stronger interac- tion among themselves [30]
Super-poster	The top 5% of the participants	Their contribution is of high quantity and quality compared to typical par- ticipants (more votes, less orphaned threads, less response time) [28]
Active participants	Write at least one forum post, read several forum posts, or do at least one quiz.	45% of them completed the course [44]
Active participants	Posted, com- mented, or voted in MOOC forums.	30% of them complete the course [32]
Active users	Constantly partici- pate in the forums for at least 6 weeks (continuance of participation not quantity)	This group of users are influential (reflected by the high number of views and votes their posts received) [45]
Representa- tive learners/ influential learners	Attract other learn- ers to interact with them. Their threads are popu- lar.	5% of the learners. High level of participation in multiple topics, above average performance, did not vote the most [46]

can be reciprocators, networkers, broadcasters, and invisible [23]; reciprocator made up the largest proportion. An interesting observation is that some participants choose to engage anonymously (about 10 percent of the posts), and the proportion of anonymous messages increases as the course proceeds [28], which would be interesting to investigate.

Participants leave footprints as they interact in discussion forums. These traces are used by researchers to understand and examine the retention, learning outcome, and interaction. Table III summarizes the relationships between the studied variables and the dropout/retention/completion. The examined variables include: types of participation (viewing, posting, volume of posts, received a reply/resolved, follow, up-vote), and content of participation (the language effects on the completion).

Previous works have studied MOOC discussion forums to understand the potential factors that may contribute to learner dropout. In fact, dropout was the most studied variable in MOOC forum participation. However, looking at the motivation of MOOC learners, dropout should not be the metric for success for the whole group of learners. It would be a suitable metric for learners who intend to complete the course as

TABLE III
THE RELATIONSHIPS BETWEEN THE STUDIED VARIABLES AND
DROUPOUT/COMPLETION

	Variable - outcome (relationship)	Method	Reference and Notes
Participation type	Posting -- passing (+)	Spearman's rank	[47]
	#posts -- passing (n)	Spearman's rank	[47]
	First cohort -- dropout (-)	Survival analysis	[29], [48] *33-35% less likely to dropout
	Visited -- retention (+)	Mann-Whit- ney	[49] *on average stay for longer period in the course.
	Receive a reply/ thread resolved -- completion (+)	Survival analysis	[50]
	Following someone -- completion (+)	Percentage comparison	[51] *completion: 50% of course activity
Participation content	Up-vote -- completion (+)	Logistic re- gression	[52] *completion: 80% of lectures are watched.
	Course-related posts -- completion (y)	paired t-test (p<0.05)	[53] * completion: take the final exam
	Cognitive engagement - dropout (-)	Survival analysis	[54]
	Expressing motivation -- dropout (-)	Survival analysis	[54], [55]
	Expressing or exposed to confusion -- dropout (+)	Survival analysis	[54], [50]
	The use of target lan- guage -- passing (n)	Spearman's rank	[47] *in Language learning MOOC
Others	More personal interpre- tation language -- drop- out (-)	Survival analysis	[55]
	Assign learners to groups based on the communication prefer- ence -- completion rate (n)	Chi-square	[56]
	Receiving encourage- ment emails (neutral tone) -- attrition (+)	Regression models	[57]
	Instructor active inter- vention -- completion rate (n)	Two-tailed z-test (p<0.05)	[26] completion can be achieved by completing one of the three op- tions: quizzes, forum badge, and project badge.
	Reputation system -- re- tention (n)	Mann- Whiteny	[49] * retention measured by FCCR

TABLE IV
THE EFFECTS OF PARTICIPATION ON LEARNING OUTCOMES

	Variable - outcome (relationship)	Method	Reference and Notes
Performance (Final Score/Exam)	Participation -- final score (+)	1. Mann-Whitney 2. Kruskal-Wallis test	1. [59] 2. [25] * Participation in terms of explicit engagement
	Participation (visits) -- (getting zero) in final score (-)	Mann-Whitney	[49] * Students who don't visit forums significantly higher in getting zero than those who visit.
	Super-poster -- grades (+)	Pearson Correlation	[28]
	More responses than questions -- grades (+)	1. Mann-Whitney 2. comparing means of two groups (T, $p < 0.01$)	1. [59] 2. [60]
	Reputation system -- grades (n)	Mann-Whitney	[49]
Quiz Score	Encouragement emails -- quiz score & quiz completion (n)	Average treatment effect	[61]
Assignment	# posts (social connections) for a certain week -- completion (score) for the accompanying assignment (+)	1. logistic regression 2. ERGM	1. [62] 2. [39]
Knowledge Gain	Level of cognitive discussion engagement -- knowledge gain (+)	Regression	[63] *Using ICAP framework

opposed to the ones who are not interested in earning a certificate. This is confirmed by the observation that only one third of learners who remain active until the end of the course will successfully complete the course [32], [47].

Another issue is the way completion is operationalized. Some studies consider passing the course to be achieved when the learner watches a certain percent of the lectures, completes a certain percentage of the course activity [58], or takes a final exam [53]. Another way to calculate the completion rate is by using First-Class-to-Course-Completion-Rate (FCCR) is a metric devised by Russell which measures the ratio of the number of students who received a passing grade to the number who viewed the first video lecture [49].

The engagement behaviors that showed increased likelihood of persistence include: posting [47], viewing [49],

receiving a reply [50], following someone [51], receiving an up-vote [52], less expression of and being exposed to confusion [54], [50], and being in the first cohort [29], [48].

The number of votes the participants receive is the most indicative feature for successful learners [52], and vague responses are negatively correlated with up-votes [24]. In addition, 20 percent of those who posted at least once received a certificate (scored 70 or more on course assignments) [39]. An interesting area to research is *What about those who received the certificate but did not contribute to the forums? Do they use forums passively? Are they already experienced with the course of study?* Anderson *et al.* [59] showed that students who received a perfect score read less on the forums than those with lower grades. However, this needs further exploration.

Conversely, factors such as assigning learners to groups based on their preferred communication mode, instructor intervention, the use of the target language, expressing cognitive engagement or motivation did not show an improvement in completion rate in their respected studies.

Participation in MOOC discussion forums indicates a correlation with performance (grades, assignment completed, and knowledge gains). Table IV shows the investigated variables and their associated with performance/learning outcome. The metric used to measure learning outcome and performance is of different granularity. The final grade is not the most accurate metric because learners may be interested in only a few topics, so assignment grades might be more accurate in this case. The factors investigated include: posting, viewing, volume of posting, type of contribution (responding or asking), level of cognitive engagement, and social connectedness. Although social connectedness in a given week seems to influence the assignment performance corresponding to that week, the findings only held true in weeks 2-4 but not beyond or before that [39].

At the end, the result suggests that there is a correlation between participation in the forums and completion and performance.

3) Efforts to Increase Participation Activity in MOOC Forums: Since the studies endorse the effects of participation on completion and performance, many researchers focused on proposing tools and methods to increase the number of participants and the volume of participation. Encouraging more active participation will help cultivate a greater sense of community, support and collaboration. Moreover, an increase in the number of participants and participation may mitigate shortcomings such as the considerable amount of unanswered questions (about 10 percent of the questions remain unanswered [32], 10-50 percent orphaned threads [28]) and the struggle of later cohorts to engage with active participants in the forums [48]. In addition, receiving a response especially to confusion not only increase learners survival [50] but also can change their attitude towards the course [53].

Some mechanisms that were developed or proposed to increase the participation activity in MOOC forums can be categorized into three types: design tools, external intervention,

TABLE V
THE RELATIONSHIP BETWEEN THE STUDIED FACTORS AND LEVEL
OF FORUM PARTICIPATION ACTIVITY

	Variable - outcome (relationship)	Method	Reference and Notes
Different mechanisms	Reputation system -- {# of responses, time for first response} (+)	Mann-Whitney	[49]
	Reputation system -- help requests (-)	Fisher's exact test	[49] less likely to post
	Up-votes -- continued participation (+)	Observation	[24]
	Staff posts -- participation (+)	Regression	[33]
	Staff post -- participation rate over time (-)		*participation: learners' post
	peer-graded assignment -- participation (+)	Regression	[33]
	#peer-graded assignment -- participation rate over time (-)		
	Receiving encouragement emails to registrants -- participation (n)	Regression	[57]
	Receiving encouragement emails -- participation (y)	Regression; Average treatment effect	[44], [61]

and recommendation techniques. Each type was assessed, although in most cases were not deployed and verified in a running course. Table V shows the relationship between investigated variables and activity level in the forums.

Design tools aim to promote incentives for learners to participate in the discussion forums. The tools were derived from successful Q&A sites [64] and include: badges, a reputation system, and fostering small sub-network communities based on space and time [65]. In [49] experiment, the use of reputation systems seems to increase the level of participation but not the final grade or completion rate. Similarly, the use of a badge system showed an increase in learners' participation and social learning [59]. The study added that the way badges are presented to the participants influences the engagement level (ex. badge-ladder has the most significant effect). Voting has a negative effect on the number of help seekers selected in a developed reputation system called "Quick Helper", but this effect is mitigated when badges that show the help givers' activity level is incorporated in the system design as well [66].

External interventions proposed were instructor intervention and sending emails to increase the engagement level. Instructors' active participation in the forums demonstrated inconsistent results. For instance, Tomkin and Charlevoix [26] did not find any significant impact on the overall completion rate, participation rate or attitudes. Moreover, there was a significant correlation with decline of participation over time [33]; it is not clear though if they control for other variables as it is generally the overall activity in the forums declines over time. At the same time, instructors' participation showed an increase in the volume of participation on the same day [33], an increase in the chance of reading the threads, and a

decrease in the negative responses [67]. Alternatively, sending self-check messages such as "Have you participated in the discussion forums?" showed increase in the participation activity [44], [61] but not the performance. Furthermore, sending encouragement emails to registrants to motivate them to participate in the forums was, nonetheless, ineffective [57]. However, sending targeted emails to participants later in the course appeared to increase the contribution when the reminder was written in a neutral tone; interestingly though it increased the likelihood of attrition [57]. Using a collectivist tone did not seem to encourage participants, which may divulge how MOOC learners perceive MOOC forums (i.e., no strong sense of community) [57].

Another approach to increase the participation and encourage social learning is to recommend potential threads of interest to potential participants or to recommend partners or peers that could respond to help seekers. Yang *et al.* [68] proposed a model that recommends a thread for participants based on three dimensions: content level, forum activity, and social connections. For example, [69] proposed a model that directs learners to threads of interest considering the level of knowledge and the balance load that learners can handle. In [20], proposed a model that recommends post feeds to participants based on their propensities to answer or seek information posted by another participant. Another work proposes partner recommendation using content and social analysis [38]. Recommendation empowers an optimal social learning environment [20]. Although many of these proposed recommendation techniques seem promising and outperform baselines, they need to be deployed and validated in real settings.

4) Analysis of Participants' Contribution in the Forums:

Inspecting the content of the participant's contribution would provide a finer grain of the learning processes, knowledge gain, and possible causes of the dropout. In the current state of MOOCs, forums are text-based, which means it is time-consuming to analyze, and in MOOC, specifically, it is overwhelming for participants to keep up with the new posts. Hecking *et al.* [70] tracked the evolution of topics and found that topics shift over time, and they may split into more than one sub-topic or evolve to a more general topic. Researchers have developed some coding schema to help categorize MOOC posts. For example, [71] categorized posts into two dimensions based on the topic and poster-role. Topics of a post can be content, other courseware, social/affective, course website, course structure, missing data, or non-English, and the poster role can be help-seeker, help-giver, or other. In [72], several subcategories were added to the poster role dimension such as complainer, sharer, or peace-maker. In addition, the authors proposed two other dimensions: the quality of the post, and the reciprocity in the thread discussion. Post messages also were categorized based on the speech act into the following: question, answer, clarification, positive feedback, negative feedback, and off-task [60], and [62], [73] included the category of issue as well. Another study classified the messages as content-related posts vs. non-content posts [74].

An analysis of the content in MOOC forums reveals that the number of content- and non-content threads are equal in proportion [74], if not in favor of content related posts [72], [75], [76]. In addition, the majority of contributions are of the question and answer types. Participants ask questions and provide answers evenly; however, the substance of the posts does not reveal meaningful discussion [72], [73]. Course content is discussed for a longer period [25], [36], [41]. Coetzee *et al.* showed that about 70 percent of the questions are content-related (41 percent regarding assignments and quizzes, and 31 percent course topics), and the rest was mainly administrative [49]. It is important to be cautious when interpreting such observations, as the context of the study may affect the results. For example, in [72] study, participation in the forums was required. Studying information discourse showed that the second post in a forum thread normally is a direct response to the first post [77]. Moreover, most of the comments seemed to be related to the parent post. However, in many cases, pairs of posts in a thread are not related, which is contrary to [78] that stated posts which appear in the same thread are mostly related to each other and can be treated independently.

Language usage varies between different groups of participants. High performers and influential participants have different styles of language discourse, which suggests they might not be the same group of learners [79]. Participants who perform well in the course engaged in “more expository style discourse with referential and deep level cohesive integration, abstract language, and simple syntactic structures” [79, p. 255], while learners who hold central positions in the social network engaged in “narrative discourse with lower referential cohesion, abstract words and simple syntactic structures” [79, p. 256]. Moreover, the language used in the forums can be a predictor for student attrition [55] and performance [59]. For example, when participants express motivation and excitement about the course [54], use first person pronouns and cognitive indicators such as thinking words (e.g., need and apply) [55] or discuss course content [53], [54], then the participants are less likely to dropout. At the same time, the language of low performers (measured by the average number of assignments submitted) was different from the high performers group; low performers tend to use non-English words and engage in study group forums [59].

Another area of research focused on studying the cognitive learning occurs within MOOC forums as a social learning environment. Wong *et al.* [42] utilized Anderson and Krathwoh’s taxonomy and found that cognitive learning terms increase in the discussion forum as the course proceeds. Dowell *et al.* [43] showed that learners’ language developed in discussion forums over time (i.e., they used more complex linguistic features measured using Coh-metrix, and Flesch-Kincaid reading level measures). Others used the interaction analysis model (IAM) to assess the cognitive level of participants’ contributions [23], [34]. Both stated that the majority of the posts did not demonstrate high level knowledge construction (did not exceed phase 3-negotiation and construction). In addition, most of the dissonance (phase 2) was noticed in the weekly modules forums [34]. Wang, Wen, and Rosé [63] evaluated the ICAP

framework claim, which asserts that the learning gains increase as the participant progresses in the cognitive engagement level (Interactive>Constructive>Active>Passive); in the study, the claim held true when Interactive and constructive levels were combined in one high-order thinking level. Hecking, Chounta, and Hoppe [37] used the same ICAP framework and found that collaborative knowledge construction and higher-order thinking occurred more frequently in smaller classes.

Examining the affective processes in MOOC forums is another topic of interest. [42] studied the affective processes over the course period and found that positive emotions decrease over time, while the tendency for negative emotions increases. [75] examined the influence of expressing or being exposed to negative sentiment to participants’ attrition. Although the results were not consistent in all examined courses, 2 out of 3 showed a positive relationship between the collective sentiment expressed in a week and the dropout rate during that week. [80] showed that negative emotions in the forums are not exclusively expressed by disengaged learners (used Opinion finder), since active participants also expressed negative emotions.

5) Efforts to Help Participants Monitor the Discussion Forums: The posts pour into MOOC forums in hundreds or thousands on average during the course period, which can easily become chaotic and overwhelming for both learners and instructors. Threads can be fragmented [81], orphaned, misplaced and noisy. Furthermore, instructional staff-to-learner ratio is highly skewed (there are a lot more learners than staff), which hampers instructional staffs’ ability to keep up-to-date with the high volume of messages. Content analytics can help alleviate these problems. Thus, another line of research in MOOC discussion forums focuses on organizing the content to assist in monitoring the forums, identifying critical posts, understanding the learning processes occurring in this environment, and enhancing the learning experience.

There are various approaches to take to organize the content and assist in monitoring the forums in an efficient manner. Classification is a common approach for this purpose in the literature. Classification is a supervised learning technique that requires a manually labeled dataset to be used to build a classifier model (the dataset is called a golden set or ground truth). Favorably, [62] confirmed that using the majority votes from crowdsourcing websites was effective for coding MOOC posts and demonstrated good agreement with experts coding.

Many papers developed models that automatically classify the topic of the post. Rossi and Gnawali [35] developed a model that predicts the forum where the post should belong. In the study, general discussions were the hardest to classify, while group study messages were the easiest [35]. Wise *et al.* [74] developed a classification model that identifies content vs. non-content starting posts (the first post of a thread), and discovered that content-related posts have distinct features that seem to be independent of the course subject domain. The study also found that the number of views and votes were not helpful in identifying content-related posts. Although this coding schema can help instructors and learners filter content-

related posts, it did not distinguish between course procedures and content specific to learning. The issue is that one-time help-seeker posts are mainly related to technical and course logistics issues [41], and course procedures topics were found to be associated with high dropout rate [54]. Therefore, if this type of post was classified early and an on-time intervention was provided, the likelihood for completion for those participants could increase [50]. Atapattu and Falkner classified posts based on the similarity between the lecture content and the post to identify the weekly forum the post should belong to [16]. Additionally, [19] use the course domain ontology to classify the posts. Researchers proposed a broader and generalizable model to classify messages on the following dimensions: confusion, sentiment, and urgency across various field domains [21], [82]. Some of them were particularly focused on models to identify confused posts and important ones.

Towards identifying confused posts, Agrawal *et al.* noticed that a confused post usually contains a question [83]. The study took a step further and developed a model that can address learner confusion by comparing the post text with lecture transcripts and recommending a video clip for the learner that is closest to the topic of confusion. The automation of intervention in this case is valuable, but a more nuanced classification of the types of messages would help in directing the learner to the right resource. In a programming course, researchers identified threads expressing frustration and classified them based on topics into four categories: execution time, error/output, code snippets, and auto-grader. More than half of the threads that received a response had discussed one of the topics identified, which calls for a better tool that address the mentioned issues [84] (read Section III-B7).

In effort to help instructors identify threads or messages that appear to be urgent and require intervention, [78], [85] developed predictive models that used the content features and thread structure [85], in addition to posting behavior and course landmark [78]. However, the proposed models did not outperform the baseline. The pitfall of these models was the assumption that considers a thread to be of interest (i.e., require attention) if it received a response from an instructor. This assumption did not account for the variation of instructional philosophies adopted by different instructors in MOOCs and how each decides to utilize the forums [85]. In [86], messages were classified based on urgency using a manually labeled dataset and stated that their proposed approach achieved an acceptable result.

The content of participants' messages was utilized to develop predictive models for different reasons. First, predictive models will help to identify influential participants, which can assist in identifying suitable prospective community teaching assistances and assist instructional staff to favorably answer their questions, assuming they will disseminate information [46]. Secondly, it can predict the correct answer [5] and, as a result, enhance learners' experience. However, in [5], content features were not as powerful compared to the other features such as first/last answer, and information about the respondent (#votes, comments and accepted answers). Third, the models can predict the performance [59], social

centrality [79], and learner at risk (attrition) [55] as discussed in Section III-B2. Furthermore, the content of the messages (words) were the most effective features to predict the speech act category for a post. However, speech act categories were not useful in predicting assignment completion or performance [62]. Last but not least, it can also be used to identify topics that learners struggle with [87] or topics that were important but not discussed [24], which will help explore themes of interest for instructional staff to make a timely adjustment and better prepare for the next iteration of the course.

When the topics and posts of interests are identified, there are two proposed ways to present them to the targeted stakeholder. Brinton *et al.* [33] proposed a ranking model to facilitate an effective, personalized recommendation. An alternative approach is visualization. This entails a word cloud visualization to depict the most discussed topics in the forums [87], a visual analytics tool that allows instructors to monitor social interactions among the participants [88], or a dashboard-oriented view of forum posts that is efficient in monitoring issues that students discussed in the forums [4].

6) Analysis of the Interaction Between Participants: Interaction is the way to facilitate social learning and stimulate higher order thinking. However, the length of a thread in MOOC forums is not an indication of deeper conversations [59]. Therefore, researchers studied social and collaborative learning in MOOC forums to understand the behavioral dynamics between learners and to identify participants who are bridges between communities and influential participants. As a result, this would improve the orchestration of the course and promotes a peer-supported learning environment that encourages knowledge sharing and construction [37], [41], [73], [89], [90].

To foster a sustainable and robust social learning environment from a network perspective, not only should good reciprocity and transitivity be sensed in the network, but there should also be a low preferential attachment effect to allow for wider information diffusion [89]. Zhang, Skryabin, and Song [89] noticed a high preferential attachment effect in MOOC forums, but this could be due to the course context (instructors' role is respected in the Chinese culture). The structural network measures showed high reciprocity, which is an indicator of responding; however, the effect of homophily and proximity were not always consistent [23]. In addition, the social network in MOOC forums showed that about 80 percent of the participants are periphery in the network, and the majority of participants have connections with at most 3 peers [23], which means low average degree [41], and low authority score [23], [73]. The rest, regular participants, have a central effect on the sustainability of the network; they tend to have good connections within their dyads; and their activities grow over time (homophily) [30]. This distribution of participants is consistent with Community of Practice members [23].

There was no evidence of an association between rich-club learners (strongly-connected component) and learning

outcomes [58]. High performers do not interact exclusively with high performing learners, nor do they hold a central position in the network [58], [79], [91]. On the contrary, participants in MOOC forums tend to communicate with other participants who belong to a different performance group, which is important for knowledge flow [91].

Instructors' interaction with participants increased the forum activity (i.e. #posts) for that specific day; however, that study showed that instructor participation was correlated with a higher decline rate [33]. A study in [92] compared the effect of instructor facilitation on participants' posting pattern. The finding shows that high instructor involvement produces high triads and low reciprocity. Conversely, in the case of no-facilitation, super-posting activity is more noticeable with higher direct reciprocity [92]. Despite some instructional staff effort to keep up and engage with the discussion, learners felt isolated [29]. In [93], there was a gap between learners' and instructors' satisfaction regarding the level of interaction in MOOC forums; instructors are more satisfied with the interaction level. Learners reported the absence of instructors as the reason for their dissatisfaction; at the same time, instructors reported lack of time as the reason for the deficiency in interactivity.

Others studied the role of the participants socially and semantically and how they interplay and develop over time. A social-semantic block-modelling method was used in [37], [94] to uncover the relation between the participants' role (help-seeking, help-giving, and core-users) and the topics they discussed. However, social and semantic roles are not completely independent, which suggests that forum communication has a limited influence on learners' interest [94]. In other words, users' based on their roles engaged differently in the forums and showed different needs and knowledge sharing which was observed in their position in the network and the content they shared [37].

Wise, Cui, and Jin emphasized the importance of considering the operationalization of the network configuration, specifically the sample selection for the experiment (posts and users), and the tie formation mechanism adopted for the study, since it has a direct impact on the results [76]. For instance, the results will be misleading if the study aims to measure learning outcomes but includes both content and no-content related posts, since non-content interactions may distort the content interactions. Similarly, the tie formation technique affects the study results. The authors of that study reported that direct reply, star, and direct reply+star showed similar network structures. However, "total co-presence presents a danger for general use due to the disproportionate influence of large threads" (pg. 389).

7) Special Topics: Discussion Forums for Programming. Learning programming requires interactive discussion forums. The current design is encumbering. [84] suggested some domain specific features that would improve users' experience and make the environment more accommodating for learning programming. For example, the platform could develop a visual forum that allows for code execution, editing, and snapshotting and integrates anchored inline annotation.

Civic Engagement. Higher education aspires to offer an inclusive learning environment and equip its students with civic competences. Some scholars examined MOOCs civic engagement by studying the discourse in a political course [95]. There was no difference in the forums participation activity between different ideologies. They cross talk and even converge to a mutual language. In addition, MOOC forums exhibited evidence of being a self-governance community via the use of an up-vote feature. The number of votes a post receives indicates the post quality and enables the creation of community boundaries. However, these boundaries alienate the opposing voices and the minority groups, which suggest that the voting feature counteracts inclusivity [96].

IV. GAPS AND FUTURE DIRECTIONS

Aggregating the research studies in this literature showed inconsistency in the terminology operationalization and metrics that measure engagement, performance, and retention. For instance, a term such as "active participant" has been defined differently, so the samples included in the studies were different, and conflict results could emerge. A lack of consensus regarding term meanings hinders researchers from supporting and validating their research findings. Future research may set a theoretical base for MOOC definitions (i.e., completion, successful completion, engagement, active participants, regular participants, etc.). In terms of metrics, they have been embraced from conventional online courses; however, the settings and participant motivation to complete the class is dissimilar. Some learners are not interested in completing the course or earning a certificate. Thus, using the completion rate in MOOCs might not be relevant as in a conventional online class. A potential task is to find new metrics for the mixed learners based on their initial intentions to complete the course and engage in the discussion forums. Alternatively, metrics that are nuanced and can measure each module engagement and performance outcome based on the engagement with the materials and activities (e.g., assessment) associated with that module (module as a unit of analysis) could be more appropriate. [49] suggested the use of pedagogical principles such as conceptual change, metacognition and project-based learning to measure learning gains.

A common gap in empirical studies of MOOC discussion forums is lack of data accessibility. Many studies have been conducted on a single course. The subject domain, the teaching method, and the role of discussion in the course are factors that influence learner participation behavior in the forums. This gap sheds light on several perspectives. First, prospective researchers need to be precise and specific in detailing the setting of the course (for example, the name of the course, the level of instructional staff engagement, whether the participation is voluntary or compulsory, the forum under study, if only one is selected) to allow for future comparison. Second, it invites platform providers to have more open access data, which will help researchers evaluate their findings in a different setting to support external validity and generalizability. Third, current research shows correlations between investigated variables, yet

researchers wish to have a survey data from the learners to help them elaborate on the findings and find explanations to the observed correlations. More available data will yield greater field advancement. A comprehensive initiative in this regard is LearnSphere project. LearnSphere serves as a portal and repository for the research community that aims to mitigate this problem and provide access to educational data collected on digital learning interaction [97].

Studies on participation focused mainly on participants who engage actively in the forums (i.e., posted at least once). Active participation may not appeal to some learners compared to others. Moreover, Mustafaraj & Bu [32] showed that there are passive learners who use the forums to find answers for their questions or to explore the media. This asserts the importance of this group of learners and the lurking behavior (listening), which remains not comprehensively explored in the literature. How frequently do these passive participants access the forums? When do their activities increase? What are the topics that attract them? How long do they spend on the threads? How does their listening behavior impact their performance? Why do they choose to be listeners? Do they find the topics they are looking for? Earlier research [31] showed that some learners choose not to engage actively in the forums because they find it overwhelming, while others do not have time for it. Understanding the listening behavior, from clickstream data, and how all learners (even the ones who just read) navigate the forums may help improve the understanding of the learning process, the topics learners struggle with, and the value of discussion forums as a learning tool.

There are still several open avenues to mitigate the overwhelming feeling users experience in MOOC forums using data mining and content analysis [98]. First, many questions that are raised are repetitive, which indicates that either learners fail to find their question in the forums or find it easier to just post a new one. A proposed solution is to identify whether the post about to be published has previously been posted, and then route the poster to that post. This will save time for learners and instructional staff and minimize duplicated effort. Second, ranking the threads based on the latest or the number of views may have an undesirable effect. Adding features like level of agreement for ranking the forums, or the quality of the threads, may attract learners to read and participate. [99] indicated that “level of agreement” ranking in online communities was associated with “attitude change”. Quality is best described as the thread relevancy and comprehensibility to the course. Third, thread summarization for long discussions that gives a preview for the conversation might be helpful if learners were interested in looking further into the discussion and engage in it. Fourth, evaluate collaborative tagging in order to filter threads based on users’ preference.

Another potential area for future research is to promote personalized learning. In personalized learning, the goal is to equip learners with the tools needed to monitor and control their learning progress based on their background and aspirations. For instance, one could develop an adaptive interface based on learner’s knowledge and interest so that threads are ranked according to their preferences. This could increase the

value and approachability of the discussion. Moreover, such a system needs to be adapted to learners’ progress. A useful tool would be a dashboard for learners that shows progress, monitors meta-cognitive tasks, and sets a learning path. There are efforts to develop visualization for instructors but few targeted to learners.

Reputation systems with different features are used to encourage learners to participate and rate each other to build trust. In addition, the reputation that participants or content builds is helpful and used in the recommender systems. The gap in the examined research is that most of these tools have not been deployed in a real-time setting, so there is no accurate evaluation as to its effectiveness, with the exception of research on the Quick Helper tool [100]. Platform providers should make it easier to integrate tools into their environment to test and advance these tools and ultimately provide a better learning experience. Future researchers in this area should know that features interplay with each other, and a different representation of the feature may have different effects on motivating learners. Thus, deploying it and evaluating a different approach is important to obtain the desirable outcome. In addition, the research emphasized the effect of these systems on increasing the engagement pattern. It would be interesting to measure the quality of the added contributions because quality is more important than just increasing participation. In the end, the purpose of a reputation system is to build trust in the community and to ensure a self-regulated environment. A successful recommender system is one that can distinguish between posts that can be automatically answered or redirected to FAQ, posts that can be answered by learners, and posts that need instructional staff to answer them.

Instructional staff showed an interest in monitoring topics that learners struggle with, content that learners most engage with and ones they consider least interesting, the effectiveness of the teaching assistants, and how to improve the presentation of the topics [4]. This information is naturally placed in discussion forums. Thus, analyzing the posts and the attitude expressed on the forums regarding the mentioned items will provide valuable insights to instructors that will help them manage the class, make appropriate adjustments, and refine the course for future iterations. Correspondingly, a visualization tool will help instructors access the information efficiently and in a timely manner.

Many forum threads in MOOCs are informational in a question and answer format, and they do not go beyond the negotiation level in the knowledge construction framework (IAM). Understanding how the learning process takes place in MOOC forums at the micro and macro level is still an interesting topic even in the conventional online learning courses. How can one foster meaningful conversations? What makes some MOOC forums present deeper learning than others? Conversational agents [100] and instructional staff presence might play a role in facilitating the discussion, yet this has not been well explored in the literature. Small groups proved to be more interactive and promote deep learning, thereby suggesting groups for participants may help. Most of the studies focused on weekly module forums and rarely investigated study group

forums. How are groups formed? What are the factors that make learners join a group? Do they stick together and help each other in different forums? Questions regarding interaction and learning require further attention. Another research direction can investigate how discussion forums are best used to optimize learning programming.

V. CONCLUSION

This paper has conducted a systematic review of the literature on the topic of MOOC discussion forums. It covers two analysis perspectives: descriptive analysis and content analysis. The descriptive analysis aggregates the reviewed papers into five dimensions: time of the publication, stakeholders involved in the analysis, objectives, methods, and data sources used in the studies. The analysis showed that most of the reviewed articles were exploration and research focused and few were targeted directly to learners. In addition, statistics and data mining were heavily used, yet more can be examined to foster a better learning experience. Future researchers are encouraged to close the learning analytics loop and contribute to the intervention and visualization objectives. Furthermore, future researchers may look to diversify their data sources by applying their models to data from different courses and platforms to measure the reliability and external validity of their models.

The content analysis of the papers included in the literature review has provided an overview of the main issues covered by research on discussion forums in the MOOC context. This paper has identified three general areas of investigation: the factors affecting the participant's behavior and learning outcome, the content analysis of the messages and how to monitor and organize them, and the interaction between learners. It sheds light into some gaps and potential areas for further investigation that hopefully will open doors for innovation and opportunities that will advance the field and make forums a better place for learning.

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